

ALTNAHARRA, United Kingdom

WMO: 030440

Lat: 58.2878N

Lon: 4.4425W

Elev: 262

StdP: 14.56

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.2	19.9	13.2	10.9	17.7	20.0	15.1	23.7	37.9	42.8	33.0	43.5	0.8	320	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	72.8	61.7	69.0	59.8	66.0	58.2	63.5	70.1	61.3	67.1	59.4	64.7	9.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.8	80.3	66.4	58.5	74.0	64.4	56.8	69.3	62.2	28.9	70.5	27.4	67.0	26.1	64.8	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.9	25.2	21.6	DB	8.4	78.0	6.7	3.4	3.6	80.4	-0.4	82.4	-4.1	84.3	-9.0	86.8
			WB	9.9	67.5	6.5	4.1	5.2	70.5	1.4	72.9	-2.3	75.2	-7.1	78.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	37.4	37.9	40.1	43.8	47.4	52.4	56.3	55.6	52.4	47.2	41.7	37.1
	DBStd	9.01	6.77	6.42	5.91	5.70	5.62	4.96	4.45	4.19	4.71	5.55	6.38	8.52
	HDD50	2195	390	340	308	200	118	30	3	5	28	118	253	402
	HDD65	7001	854	759	772	637	545	378	272	294	377	551	698	864
	CDD50	670	0	1	1	13	37	103	198	177	101	32	5	2
	CDD65	4	0	0	0	0	0	0	2	2	0	0	0	0
	CDH74	69	0	0	0	0	7	12	34	15	1	0	0	0
	CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0
	WSAvg	8.7	9.9	9.9	10.0	9.1	8.5	8.3	7.1	7.3	8.0	8.9	8.9	8.5
	PrecAvg	44.6	5.1	4.0	3.6	2.8	2.8	2.5	2.5	3.3	3.7	4.8	5.2	4.8
(15) Precipitation	PrecMax	53.7	9.3	7.1	6.7	5.0	5.3	4.3	4.4	6.8	10.4	10.5	7.7	8.8
	PrecMin	35.4	0.7	1.2	1.1	1.0	0.1	1.1	0.9	1.5	1.2	1.3	2.5	0.8
	PrecStd	5.0	2.7	1.7	1.6	0.9	1.3	0.8	1.1	1.3	1.9	2.2	1.5	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	54.8	58.5	66.7	74.1	75.4	78.5	76.3	71.5	63.0	58.1	55.7
		MCWB	49.2	50.3	49.5	54.0	59.4	61.6	64.9	64.9	60.8	55.9	52.9	51.3
	2%	DB	51.1	52.4	53.7	60.9	67.8	69.2	72.8	70.0	66.4	60.2	54.9	53.6
		MCWB	47.5	48.9	48.1	52.0	56.1	58.7	63.3	61.6	58.5	54.6	50.8	50.0
	5%	DB	49.2	50.2	51.3	57.2	62.9	65.0	68.7	66.3	63.4	58.0	52.9	51.6
		MCWB	46.0	47.0	47.1	50.2	54.0	56.7	60.8	58.9	56.8	53.1	49.3	48.2
	10%	DB	47.0	47.4	49.1	54.0	58.8	61.8	65.3	63.8	60.8	55.7	51.1	48.9
		MCWB	44.1	44.0	45.2	48.3	51.8	55.1	59.0	57.6	55.3	51.5	47.9	45.7
	0.4%	WB	50.5	51.4	51.5	56.3	62.2	63.4	67.2	66.7	62.5	57.8	54.0	52.4
		MCDB	52.8	53.7	55.1	63.6	69.8	73.0	74.2	73.4	68.0	61.0	57.1	54.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.3	49.5	49.3	53.0	57.9	59.9	63.9	62.7	59.7	55.6	51.6	50.6
		MCDB	50.6	52.0	52.6	59.0	65.8	67.3	71.0	68.3	64.7	59.2	54.4	53.2
	5%	WB	46.3	47.3	47.5	51.0	54.7	57.8	61.8	60.2	57.9	53.7	49.7	48.5
		MCDB	49.0	50.2	50.8	56.3	61.8	63.8	67.6	65.0	62.2	57.1	52.3	51.5
	10%	WB	44.3	44.5	45.6	49.0	52.3	55.8	59.7	58.2	56.2	52.1	48.1	46.0
		MCDB	46.9	47.2	49.0	53.8	58.3	61.0	64.8	62.7	60.1	55.2	51.0	49.1
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.9	13.1	15.6	17.5	14.8	14.5	14.2	14.6	12.6	11.0	11.6
		MCDBR	10.9	13.0	16.5	22.5	26.8	21.9	22.3	19.8	20.2	14.7	11.9	12.8
		MCWBR	9.6	10.3	11.4	13.7	16.7	13.0	13.6	12.2	12.9	10.6	9.9	11.0
		MCDBR	10.6	11.1	14.2	19.1	24.0	19.5	20.5	17.9	17.0	12.8	11.4	12.2
	5% WB	MCWBR	9.7	9.8	11.2	12.8	15.9	12.5	13.2	11.9	12.1	10.5	10.2	11.1
(40) Clear-Sky Solar Irradiance	taub		0.322	0.343	0.361	0.379	0.377	0.371	0.376	0.365	0.354	0.340	0.321	0.308
	taud		2.162	2.261	2.294	2.291	2.350	2.403	2.403	2.441	2.459	2.387	2.216	2.107
	Ebn at Noon		153	213	246	262	271	275	270	266	250	215	154	116
	Edh at Noon		16	23	30	35	36	34	33	30	25	21	15	13
(44) All-Sky Solar Radiation	RadAvg		107	294	629	1093	1475	1451	1352	1094	735	377	147	71
	RadStd		14	37	48	74	102	131	102	62	51	35	15	8

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page